

Shankhanil Borthakur

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EDUCATION

National Institute of Technology, Surat

B.Tech. in Electronics and Communication (Major), Computer Science (Minor)

Aug 2019 - May 2023

GPA: 9.55/10 (3.95/4)

Coursework: Data Structures & Algorithms, Computer Architecture, Computer Networks, Operating System, Deep Learning

PROFESSIONAL EXPERIENCE

Software Engineer 2, JPMorgan Chase

Aug 2023 - Present

- Developed a cloud-based **distributed** application that processes **5 million** transactions daily, handles over **50,000** analytics queries per hour, and underpins a **\$2 billion** annual trading business.
- Engineered a **hot and cold** data strategy that uses **Query Federation** to access recent data from high-performance Redshift storage and historical data from Amazon S3. This solution slashed storage costs by **30%**.
- Designed a database synchronization utility to ingest on-field survey data into a remote PostGIS database for real-time geospatial analytics. This automated **95%** of our clients' manual workflow, saving time and resources.

Software Engineer Intern, JPMorgan Chase

May 2022 - July 2022

- Designed an Amazon Kinesis data pipeline to **stream** nearly **1 Terabyte** of high-frequency trading data each day to Amazon Redshift. This on-prem to cloud transition enabled **4X** operational scaling for one of the firm's fastest-growing trading divisions.

RESEARCH EXPERIENCE

Summer Research Fellow, Indian Institute of Technology Delhi

May 2021 - July 2021

- Investigated the application of **Twin Augmentation** in pre-trained classifiers to address **class imbalance** in chest X-ray classification within Covid-19 datasets. This method achieved a **5%** improvement in classification accuracy over the baseline model.

Research Intern, Indian Space Research Organization

Dec 2020 - Feb 2021

- Developed a **geolocation algorithm** to map Mask R-CNN instance segmentation outputs to precise geo-coordinates in satellite imagery. This work supported ISRO's research on geolocation-aware deep learning models for **remote sensing** applications.

PUBLICATIONS

- NER for Extracting Tourist Attractions from Textual Information: A BERT-BiLSTM-CRF Approach, *2023 IEEE 5th ICAECC* 
- Object Detection for Military Surveillance using YOLO Framework, *2023 IEEE 20th INDICON* 

PROJECTS

Streak Fit | AI-Powered Virtual Group Fitness Platform

- Led** a three-member team to develop a platform where users can exercise with their peers via **video conferencing** and a **pose detection** model monitors their workout movements and provides real-time feedback.

Travelo | Smart Travel Planning using Named Entity Recognition

- Built a web application capable of automatically **parsing** lengthy travel blogs to **extract** tourist destinations and generate an itinerary. The application also plots an **efficient route** connecting the travel destinations on a map.

Q&A Made Easy | Real-Time Q&A Moderation System using NLP

- Developed a Chrome extension with a built-in **NLP filter** to remove **profane** and **repetitive** questions during online meetings. Integrated **speech-to-text** support for users to post questions and an admin dashboard for **streamlined** Q&A moderation.

TECHNICAL SKILLS

Languages: Python, C++, Java

Cloud: AWS, GCP, Terraform

AI & ML: TensorFlow, Keras, PyTorch, Pandas, NumPy, scikit-learn, OpenCV

Web Technologies: React, Node.js, Django, Docker, WebRTC, Sockets.IO

Databases: MySQL, MongoDB, DynamoDB, Aurora, Redshift

ACHIEVEMENTS

- Gold Medalist**, ranked 1 out of 156 students with a GPA of 9.55/10 in the Department of Electronics Engineering, NIT Surat.
- Accelerated Promotion** to Software Engineer 2 at JPMorgan, awarded to the **top 5%** of global cohort for impactful performance.
- Winner** of **7** hackathon awards, including annual competitions at the **University at Buffalo** and **Fordham University**.

COMMUNITY LEADERSHIP

Microsoft Student Ambassador: Mentored **50+** students in web development and organized **10+** programming workshops.

Organizer, DotSlash Hackathon: Spearheaded the **technical operations**, developed event **website**, hosted **live sessions** on YouTube.